02 GROUND CANCEL BUFFER, MIX L

File: master_sheets/02_ground_cancel_buffer.kicad_sch

03 MIX L, MIX R

File: master_sheets/03_mix_l.kicad_sch

04 MIX R, MASTER FADER

File: master_sheets/04_mix_r.kicad_sch

05 MASTER FADER, MONO

File: master_sheets/05_master_fader.kicad_sch

06 MAIN OUTPUTS

File: master_sheets/06_main_outputs.kicad_sch

07 MAIN OUTPUTS, POWER-ON MUTE

File: master_sheets/07_main_outputs.kicad_sch

08 POWER-ON MUTE, AUX MASTER 5

File: master_sheets/08_power_on_mute.kicad_sch

09 AUX MASTER 5

File: master_sheets/09_aux_master_5.kicad_sch

10 AUX MASTER 6

File: master_sheets/10_aux_master_6.kicad_sch

11 AUX MASTER 6, EXT INPUT

File: master_sheets/11_aux_master_6.kicad_sch

12 EXT INPUT, PFL-AFL

File: master_sheets/12_ext_input.kicad_sch

13 PFL-AFL, PFL SWITCHER

File: master_sheets/13_pfl_afl.kicad_sch

14 PFL SWITCHER, PFL DETECT

File: master_sheets/14_pfl_switcher.kicad_sch

15 PFL DETECT, RUDE SOLO

File: master_sheets/15_pfl_detect.kicad_sch

16 CONTROL ROOM

File: master_sheets/16_control_room.kicad_sch

17 CONTROL ROOM, HEADPHONES

File: master_sheets/17_control_room.kicad_sch

18 HEADPHONES, METER L

File: master_sheets/18_headphones.kicad_sch

19 METER L

File: master_sheets/19_meter_l.kicad_sch

20 METER L

File: master_sheets/20_meter_l.kicad_sch

21 METER L

File: master_sheets/21_meter_l.kicad_sch

22 METER L

File: master_sheets/22_meter_l.kicad_sch

23 METER R

File: master_sheets/23_meter_r.kicad_sch

24 METER R

File: master_sheets/24_meter_r.kicad_sch

25 METER R

File: master_sheets/25_meter_r.kicad_sch

26 METER R

File: master_sheets/26_meter_r.kicad_sch

27 JACK LINK CLAMPS

File: master_sheets/27_jack_link_clamps.kicad_sch

28 JACK LINK CLAMPS, SPARE BUS GROUNDING

File: master_sheets/28_jack_link_clamps.kicad_sch

29 POWER ENTRY, DISTRIBUTION

File: master_sheets/29_power_entry.kicad_sch

30 DISTRIBUTION, MOTHERLOOM

File: master_sheets/30_distribution.kicad_sch

31 REMOTE METER HEADER, JACK BOARD LINK

File: master_sheets/31_remote_meter_header.kicad_sch

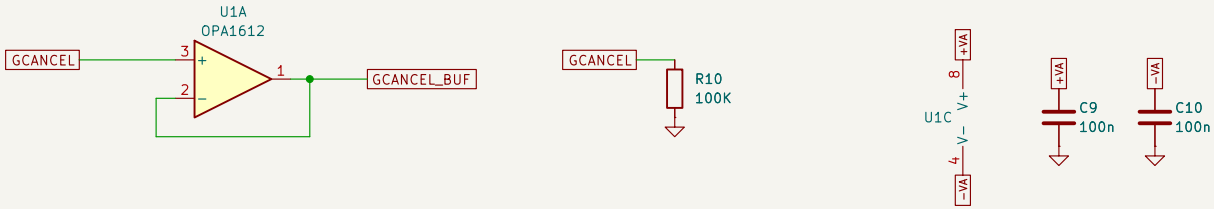
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14

Sheet: /
File: master.kicad_sch

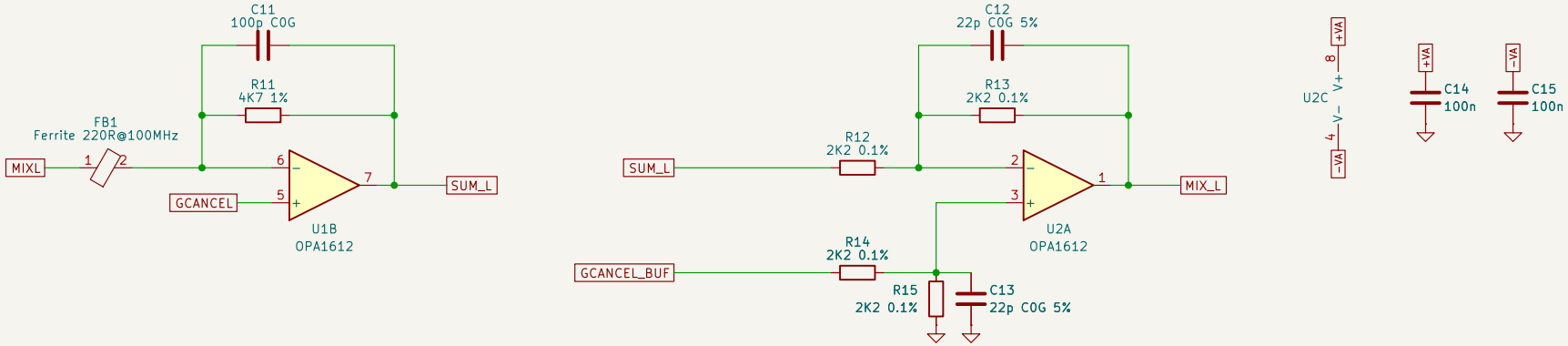
Title: Master section — goa console

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 1/31

GROUND CANCEL BUFFER (feeds every re-inverter; summing amps use the raw GCANCEL line; 100K keeper)



MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert



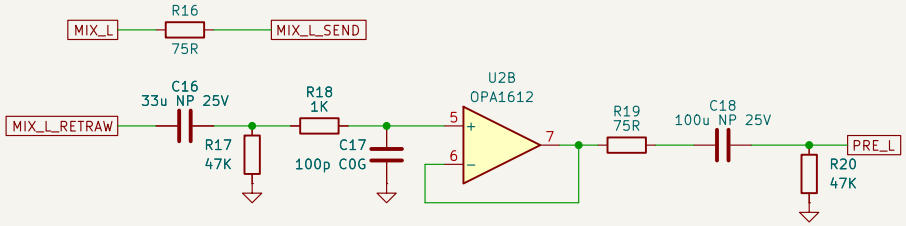
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /02 GROUND CANCEL BUFFER, MIX L/
File: 02_ground_cancel_buffer.kicad_sch

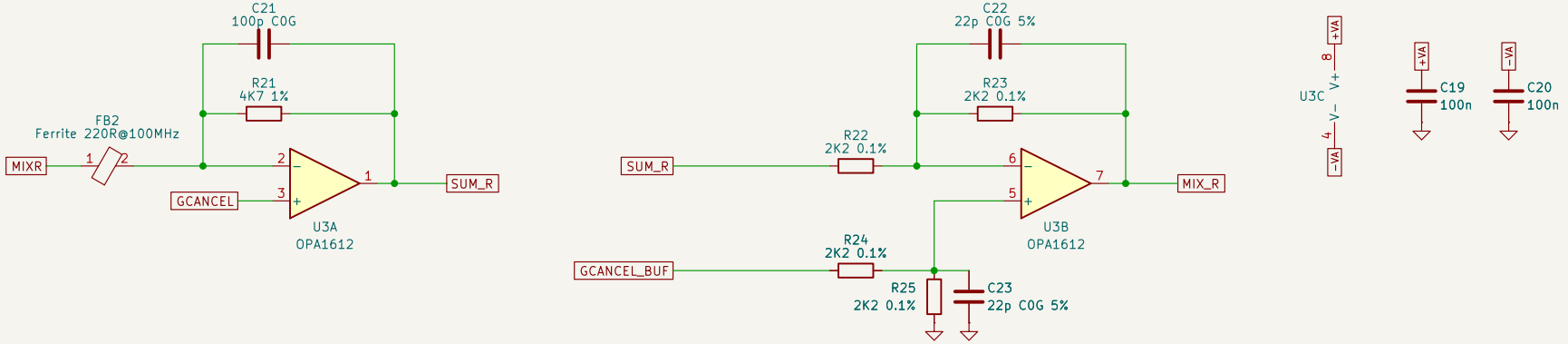
Title: GROUND CANCEL BUFFER, MIX L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 2/31

MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert (cont.)



MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert



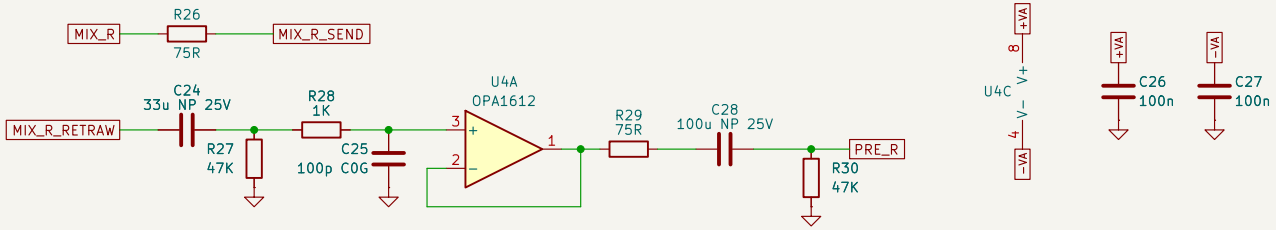
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /03 MIX L, MIX R/
File: 03_mix_l.kicad_sch

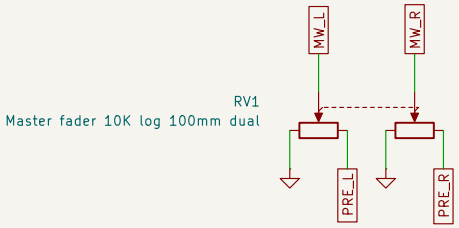
Title: MIX L, MIX R

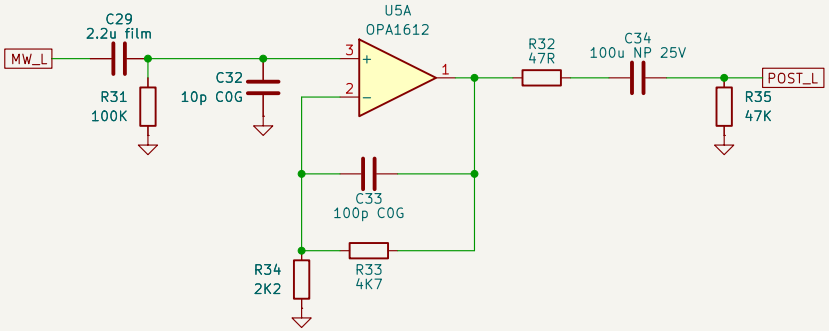
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 3/31

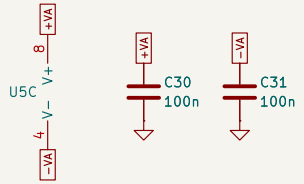
MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert (cont.)



MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB







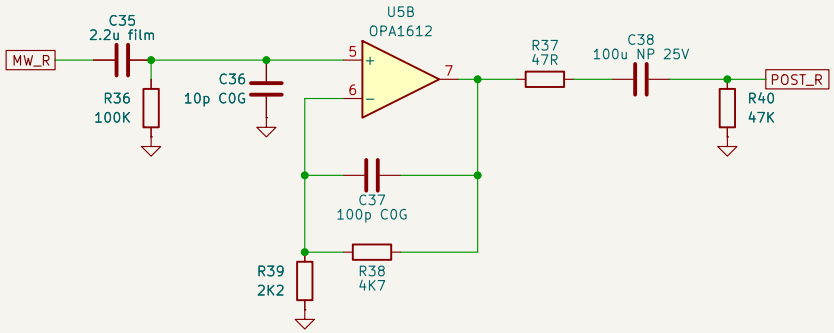
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /04 MIX R, MASTER FADER/
File: 04_mix_r.kicad_sch

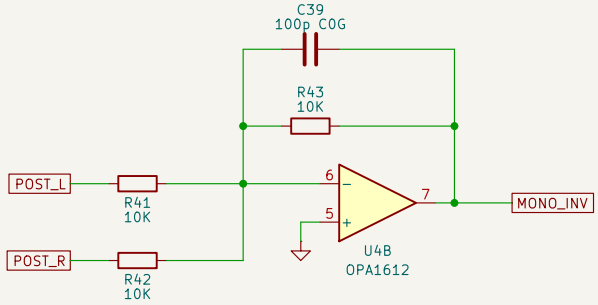
Title: MIX R, MASTER FADER

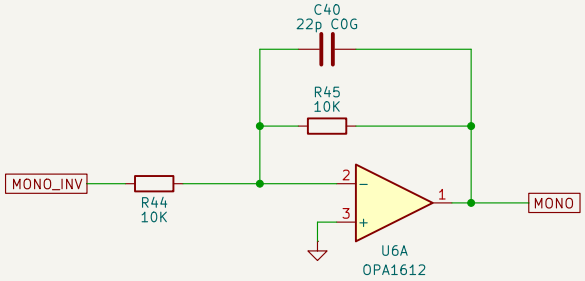
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 4/31

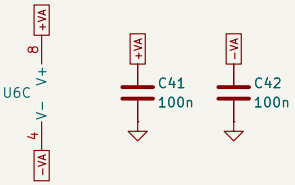
MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB (cont.)

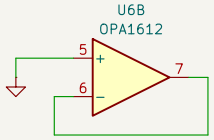


MONO: inverting sum of POST L+R, re-inverter









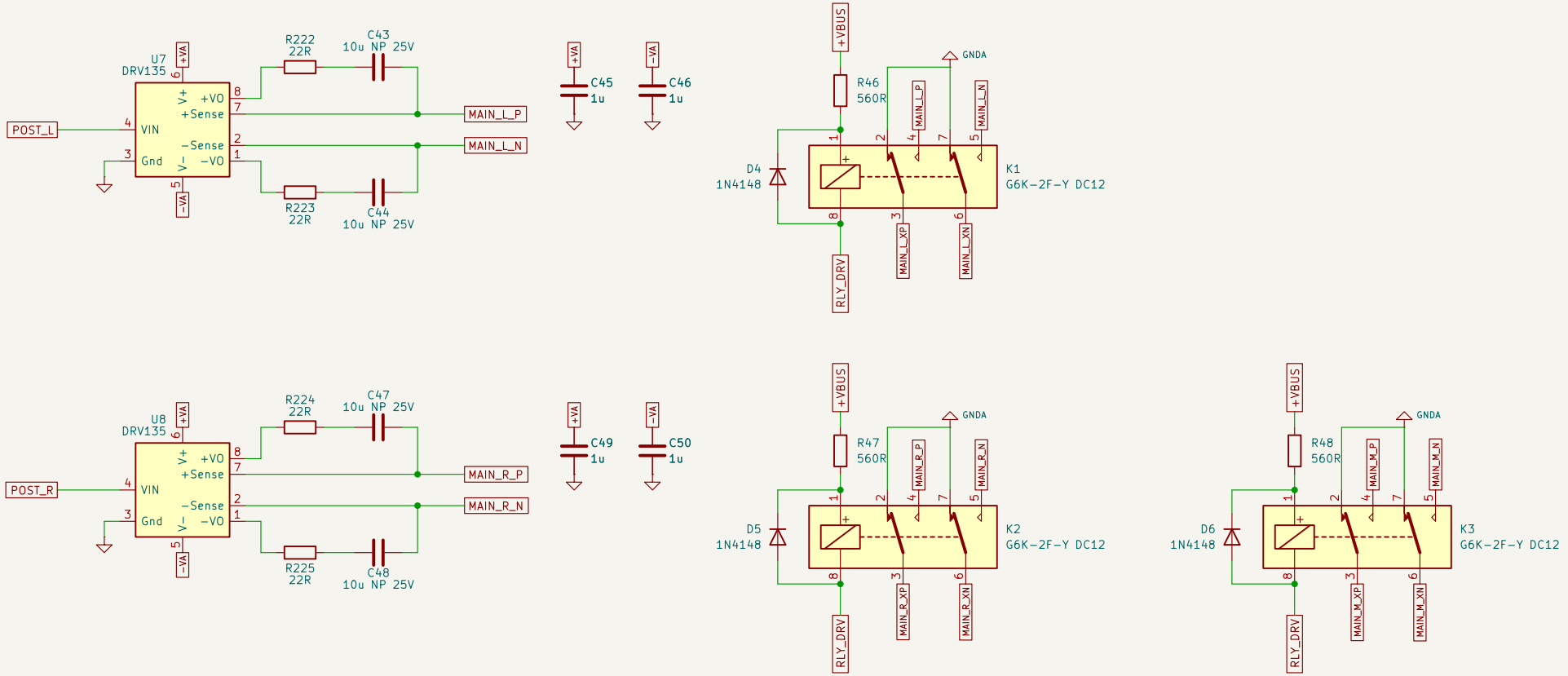
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /05 MASTER FADER, MONO/
File: 05_master_fader.kicad_sch

Title: MASTER FADER, MONO

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 5/31

MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO; NC grounds the line) -> XLR male, pin 2 hot



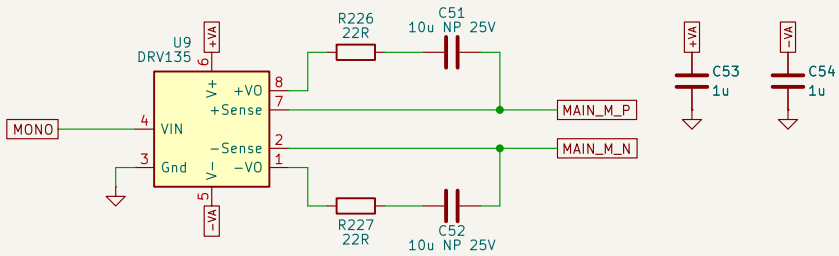
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /06 MAIN OUTPUTS/
File: 06_main_outputs.kicad_sch

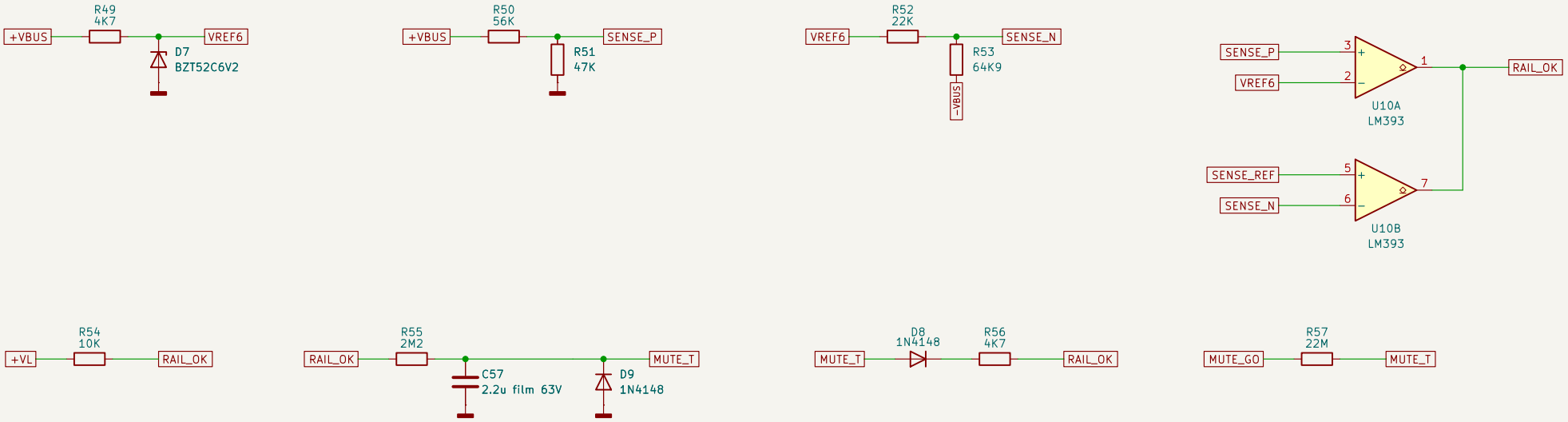
Title: MAIN OUTPUTS

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 6/31

MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO; NC grounds the line) -> XLR male, pin 2 hot (cont.)



POWER-ON MUTE: rails > ~14 V for ~2.7 s closes all six relays; either rail low opens them at once



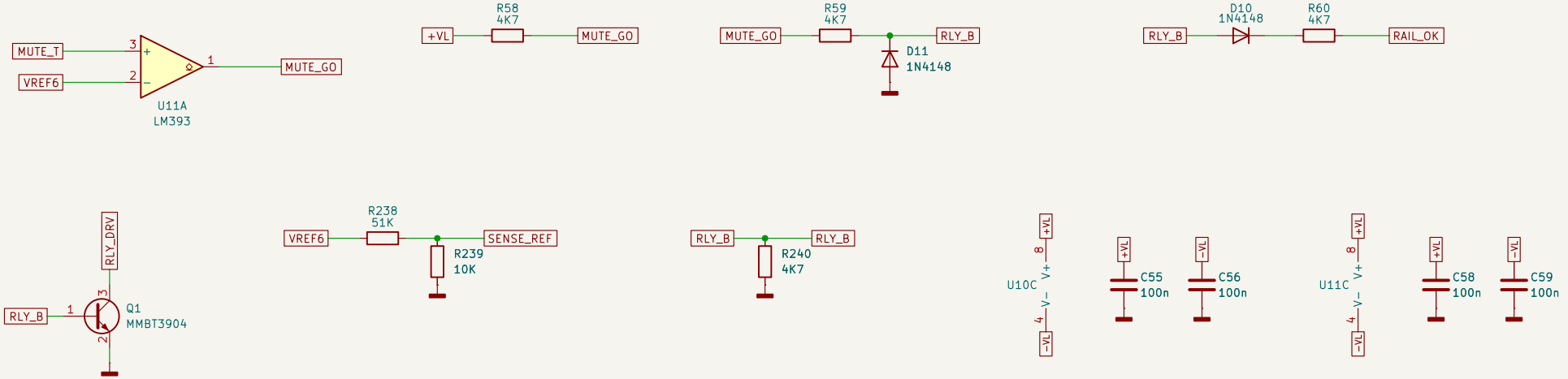
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /07 MAIN OUTPUTS, POWER-ON MUTE/
File: 07_main_outputs.kicad_sch

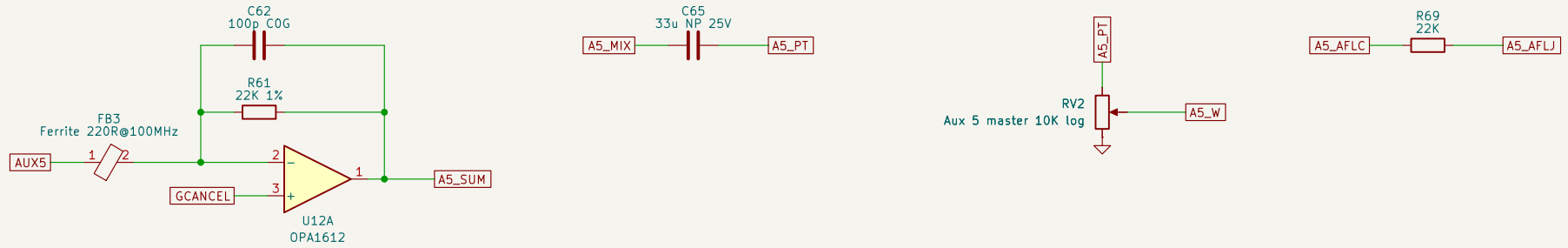
Title: MAIN OUTPUTS, POWER-ON MUTE

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 7/31

POWER-ON MUTE: rails > ~14 V for ~2.7 s closes all six relays; either rail low opens them at once (cont.)



AUX MASTER 5: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL



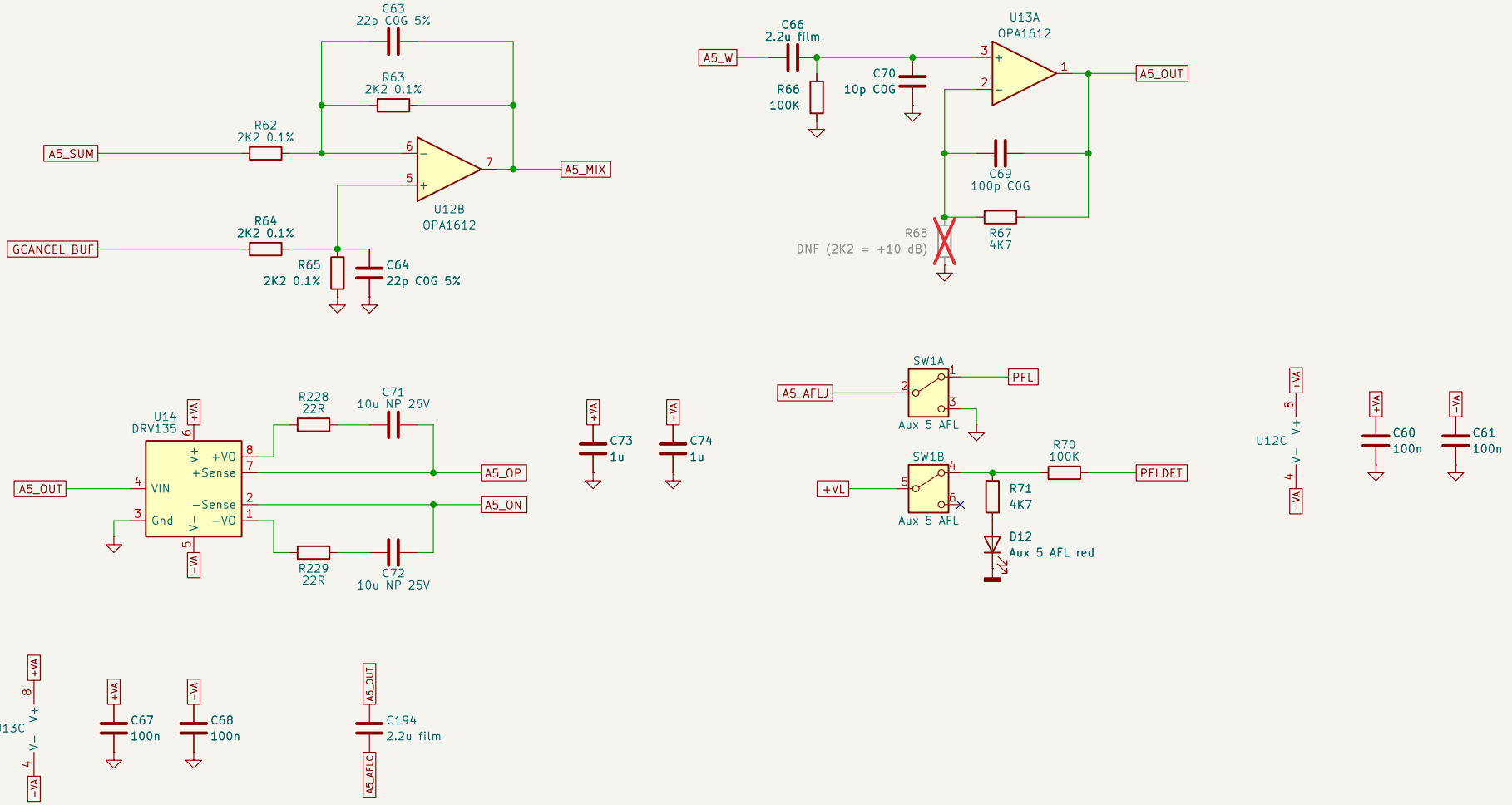
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /08 POWER-ON MUTE, AUX MASTER 5/
File: 08_power_on_mute.kicad_sch

Title: POWER-ON MUTE, AUX MASTER 5

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 8/31

AUX MASTER 5: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)



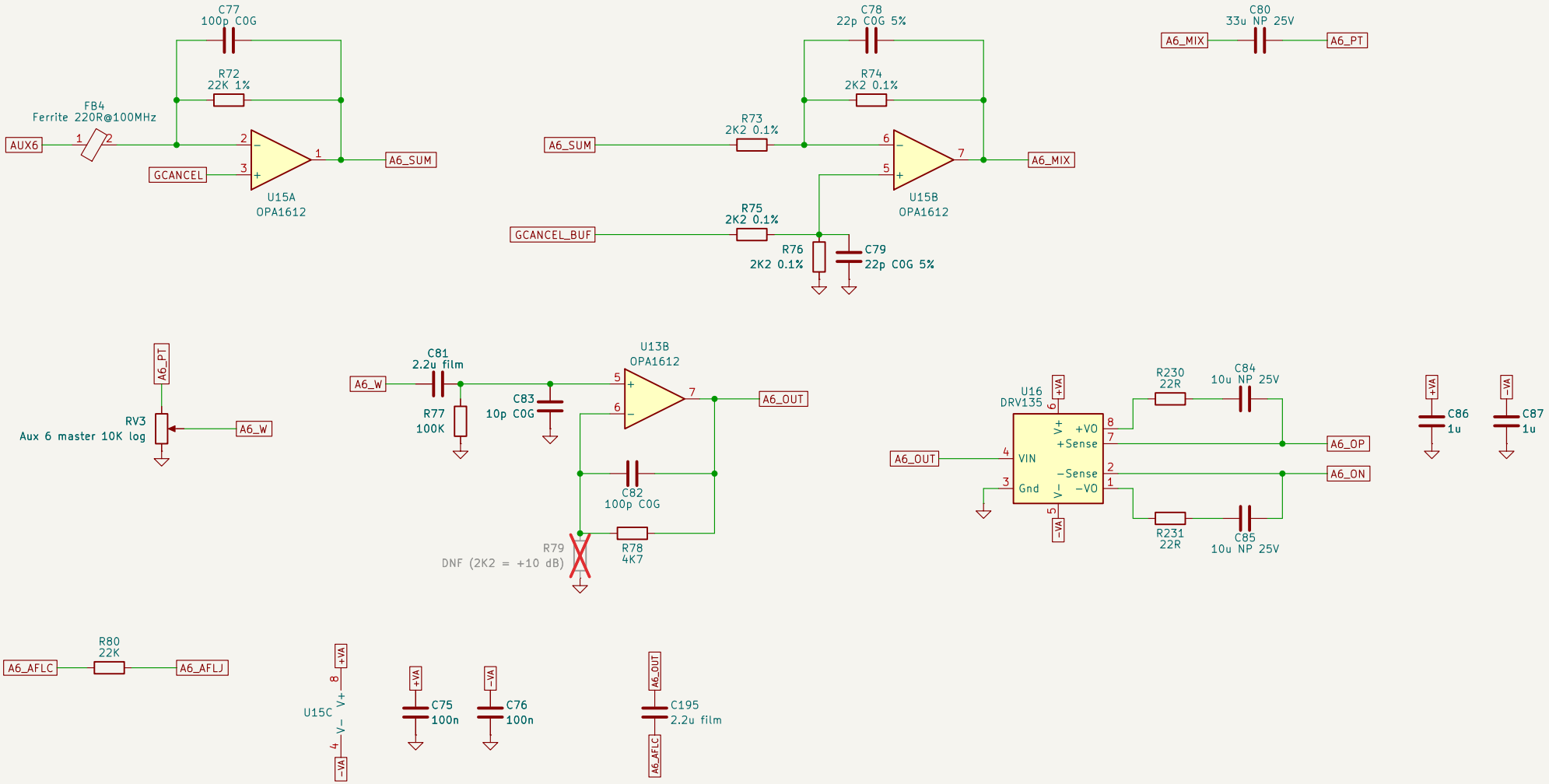
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /09 AUX MASTER 5/
File: 09_aux_master_5.kicad_sch

Title: AUX MASTER 5

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 9/31

AUX MASTER 6: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL



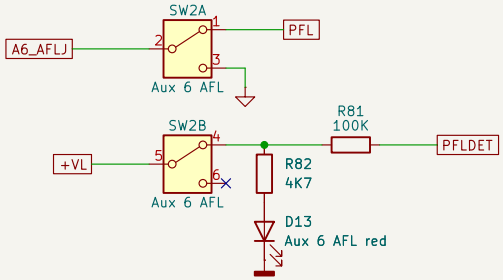
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /10 AUX MASTER 6/
File: 10_aux_master_6.kicad_sch

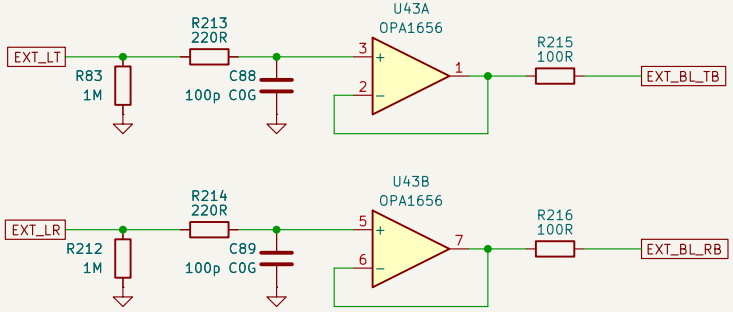
Title: AUX MASTER 6

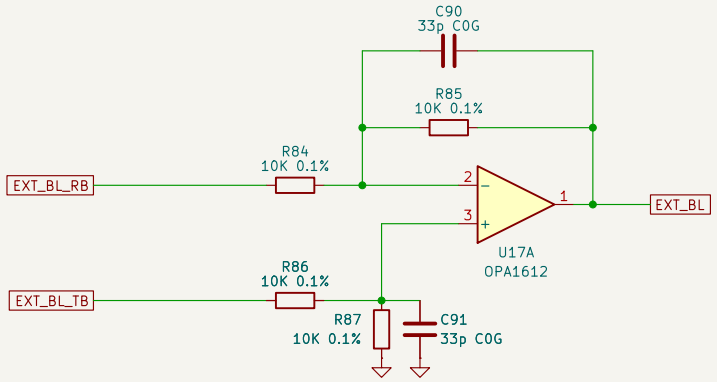
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 10/31

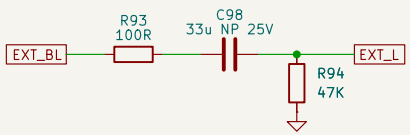
AUX MASTER 6: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)

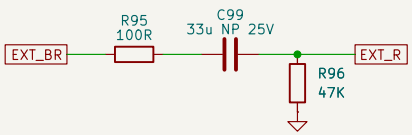


EXT INPUT (2-track / DAW) and MIX / EXT monitor select (SW latched = EXT)









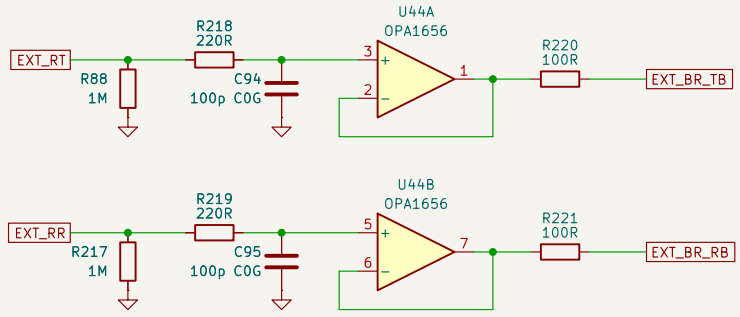
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

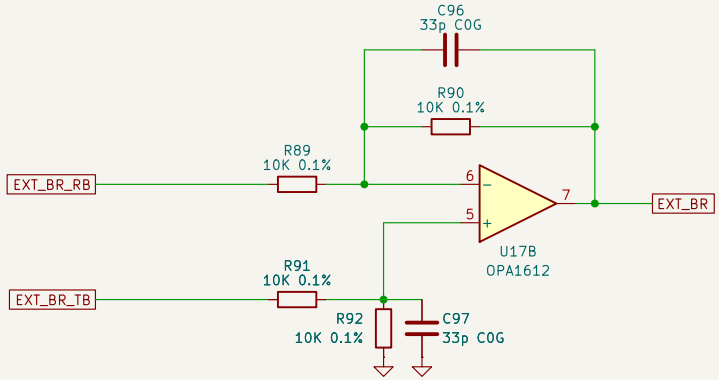
Sheet: /11 AUX MASTER 6, EXT INPUT/
File: 11_aux_master_6.kicad_sch

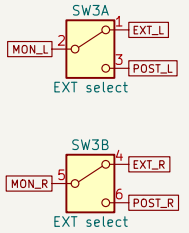
Title: AUX MASTER 6, EXT INPUT

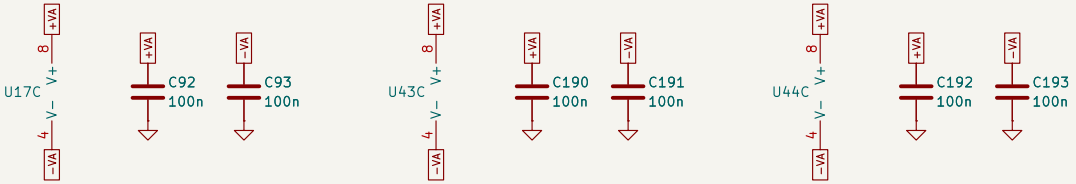
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 11/31

EXT INPUT (2-track / DAW) and MIX / EXT monitor select (cont.)

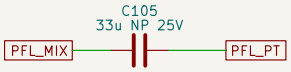


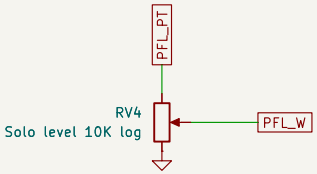


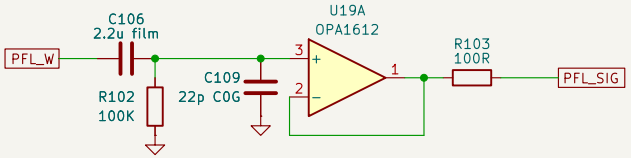


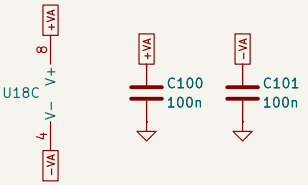


PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower









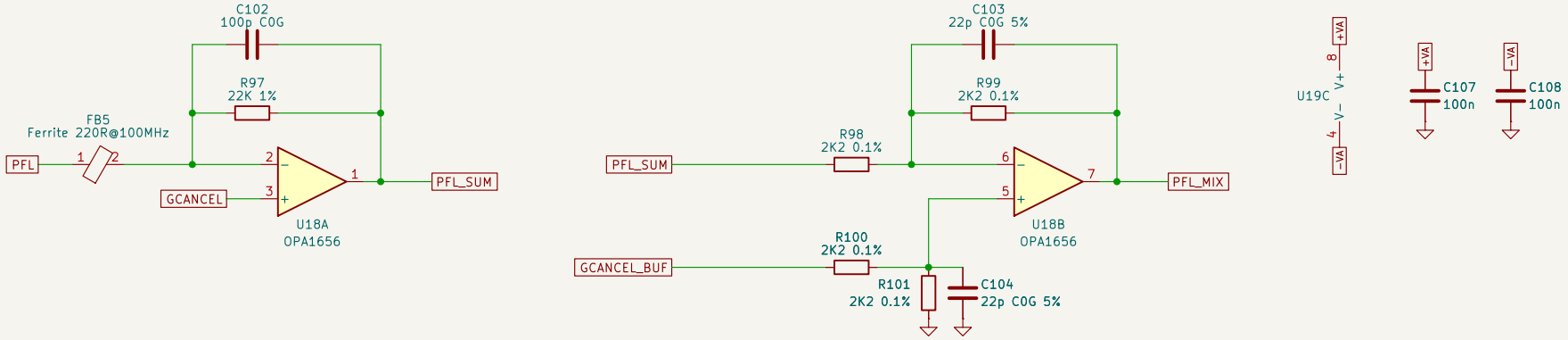
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /12 EXT INPUT, PFL-AFL/
File: 12_ext_input.kicad_sch

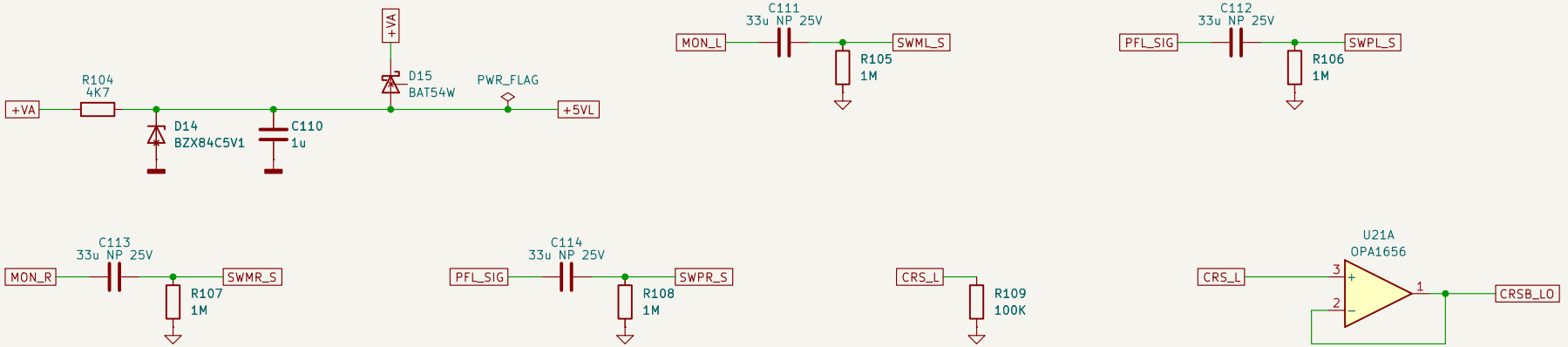
Title: EXT INPUT, PFL-AFL

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 12/31

PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower (cont.)



PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break-before-make) -> CRS L/R OPA1656 followers



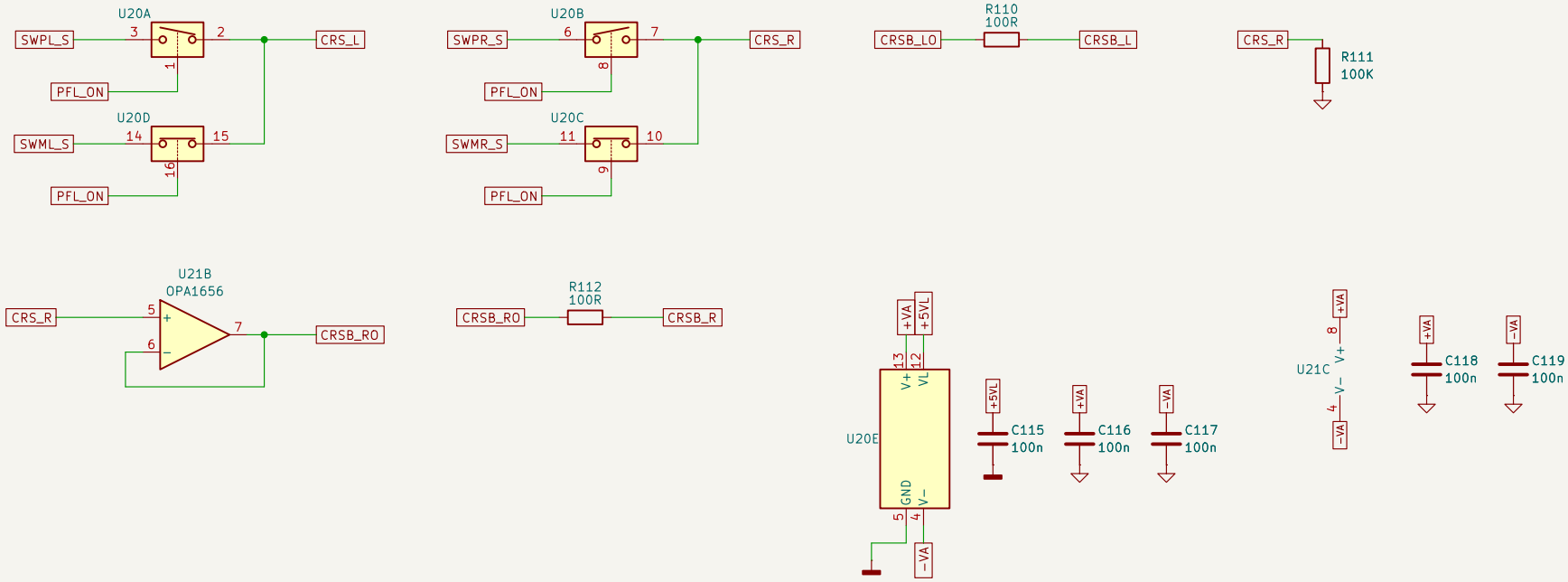
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /13 PFL-AFL, PFL SWITCHER/
File: 13_pfl_afl.kicad_sch

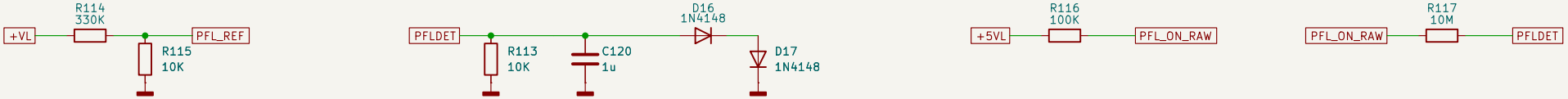
Title: PFL-AFL, PFL SWITCHER

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 13/31

PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break-before-make) -> CRS L/R OPA1656 followers (cont.)



PFL DETECT (0.92 V per press vs 0.50 V; 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher



Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /14 PFL SWITCHER, PFL DETECT/
File: 14_pfl_switcher.kicad_sch

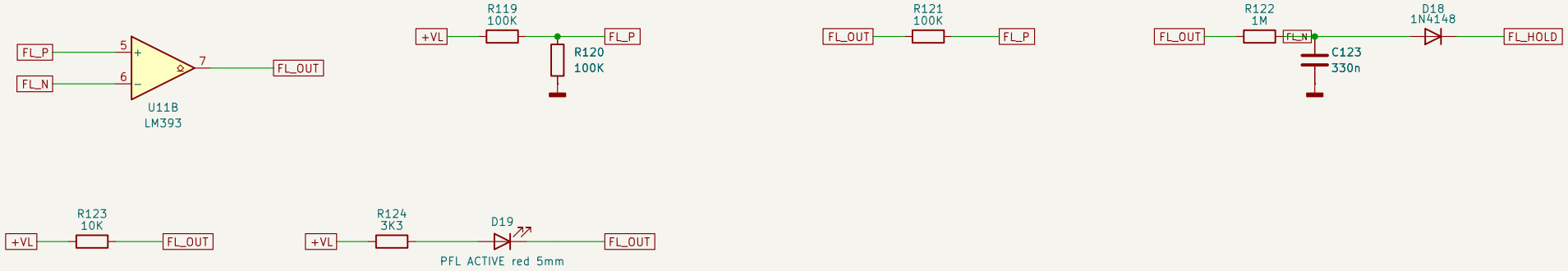
Title: PFL SWITCHER, PFL DETECT

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 14/31

PFL DETECT (0.92 V per press vs 0.50 V; 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher (cont.)



RUDE SOLO: ~2 Hz flasher, PFL ACTIVE LED; held off through FL_N while no key is pressed



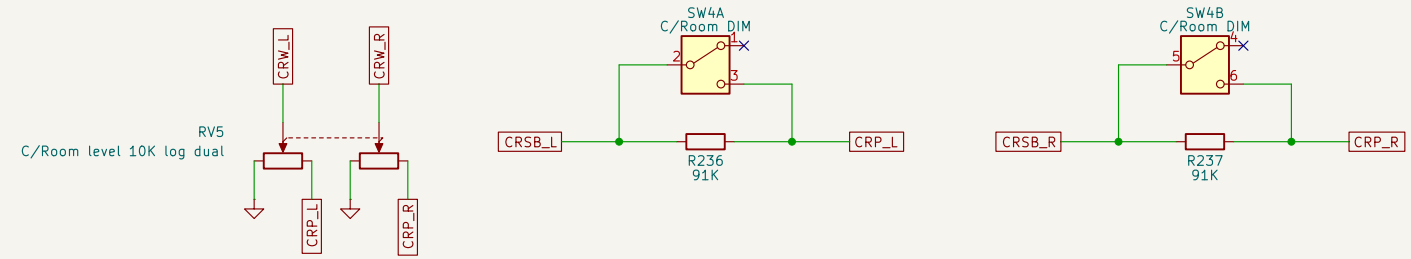
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /15 PFL DETECT, RUDE SOLO/
File: 15_pfl_detect.kicad_sch

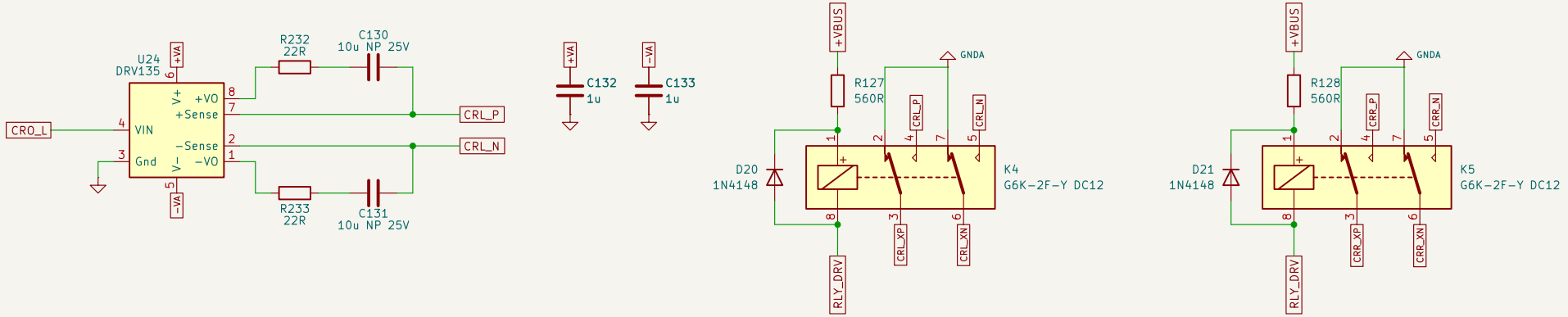
Title: PFL DETECT, RUDE SOLO

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 15/31

CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs







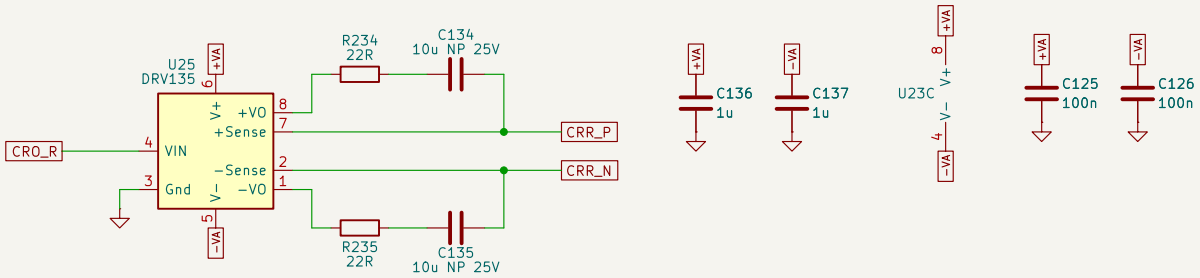
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /16 CONTROL ROOM/
File: 16_control_room.kicad_sch

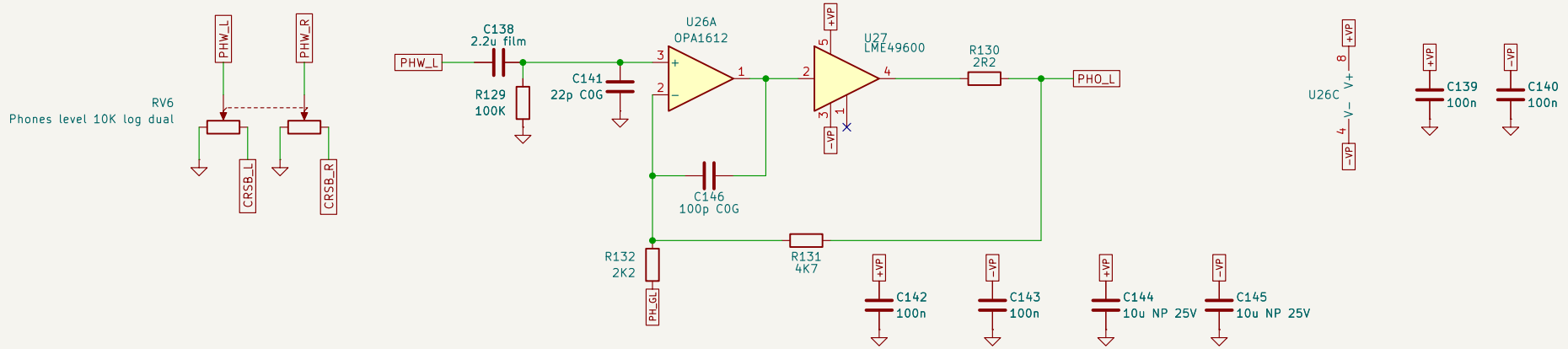
Title: CONTROL ROOM

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 16/31

CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs (cont.)



HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, mute relay, PHONESGND



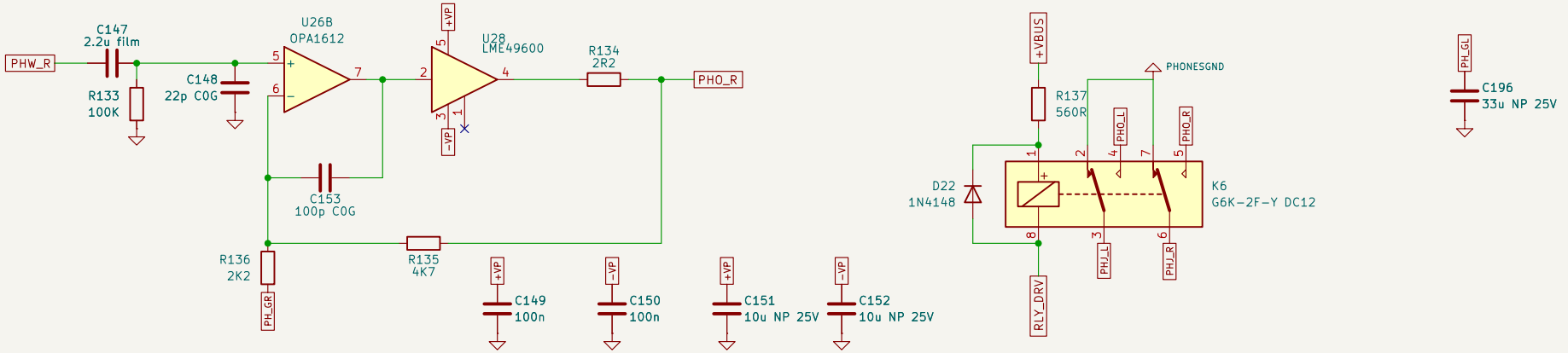
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

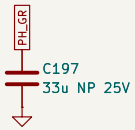
Sheet: /17 CONTROL ROOM, HEADPHONES/
File: 17_control_room.kicad_sch

Title: CONTROL ROOM, HEADPHONES

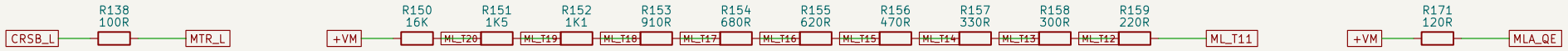
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 17/31

HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, mute relay, PHONESGND (cont.)





METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings



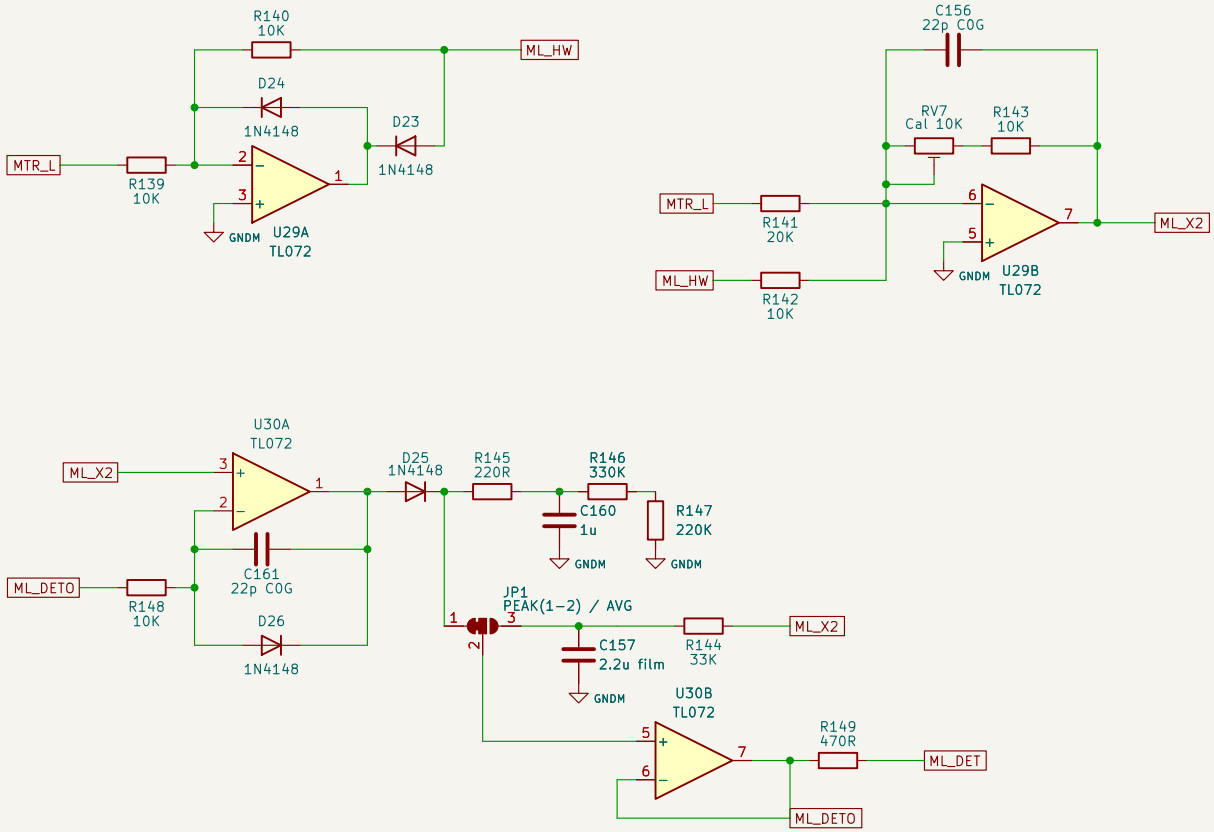
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /18 HEADPHONES, METER L/
File: 18_headphones.kicad_sch

Title: HEADPHONES, METER L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 18/31

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



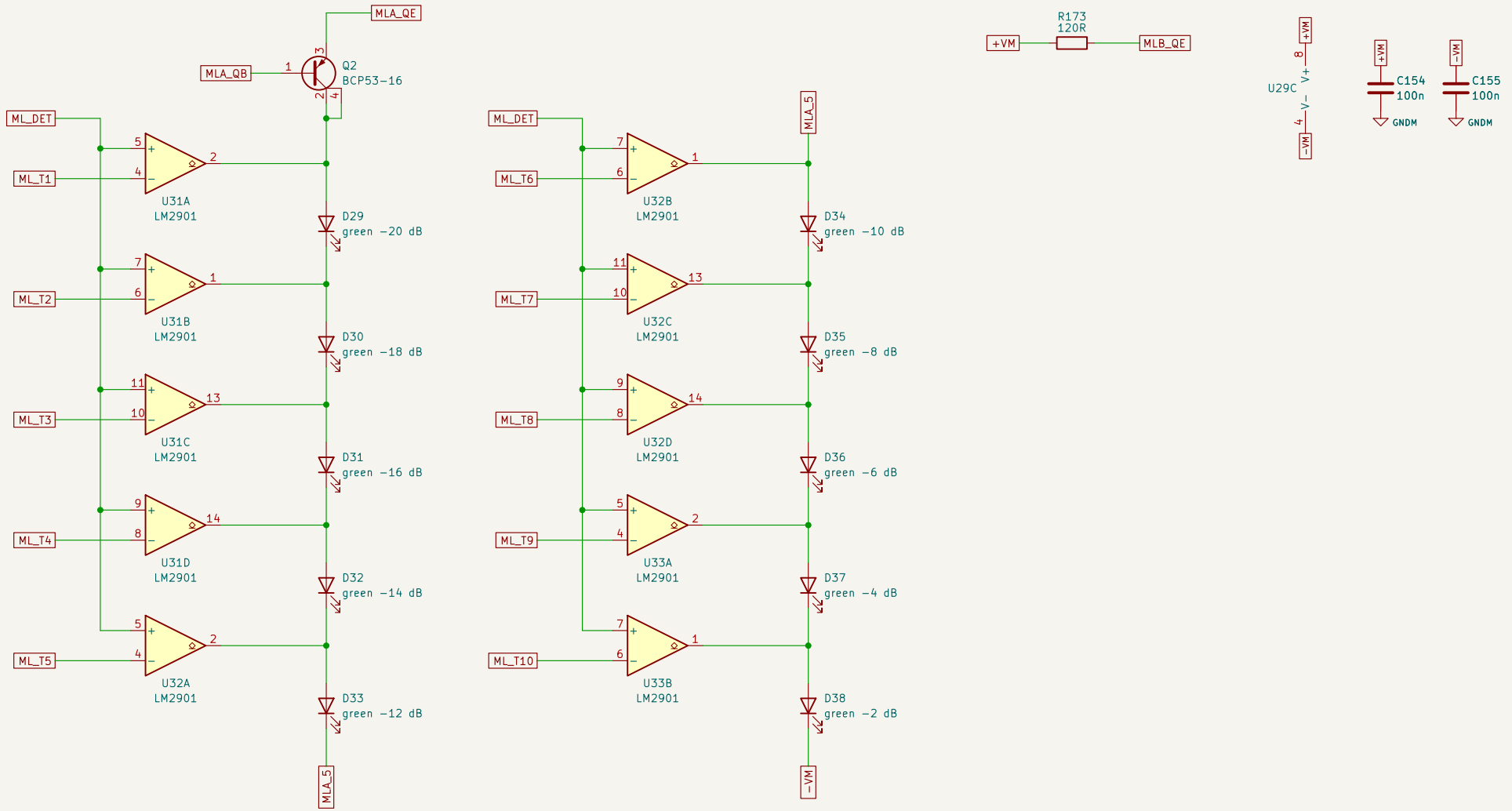
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /19 METER L/
File: 19_meter_l.kicad_sch

Title: METER L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 19/31

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



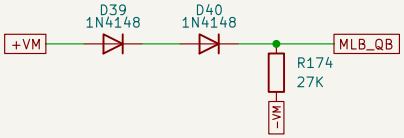
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

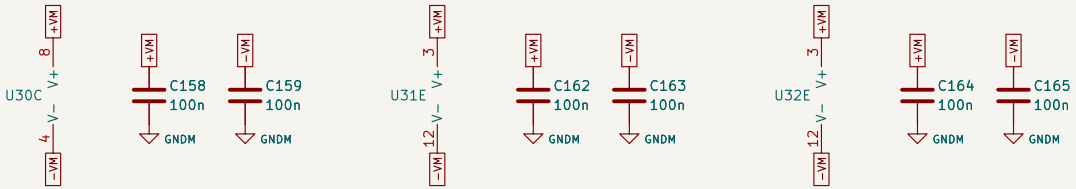
Sheet: /20 METER L/
File: 20_meter_l.kicad_sch

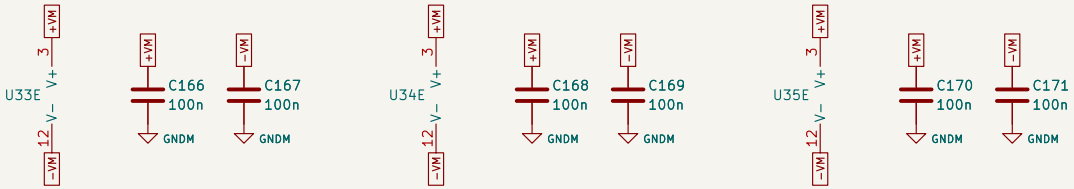
Title: METER L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 20/31

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)







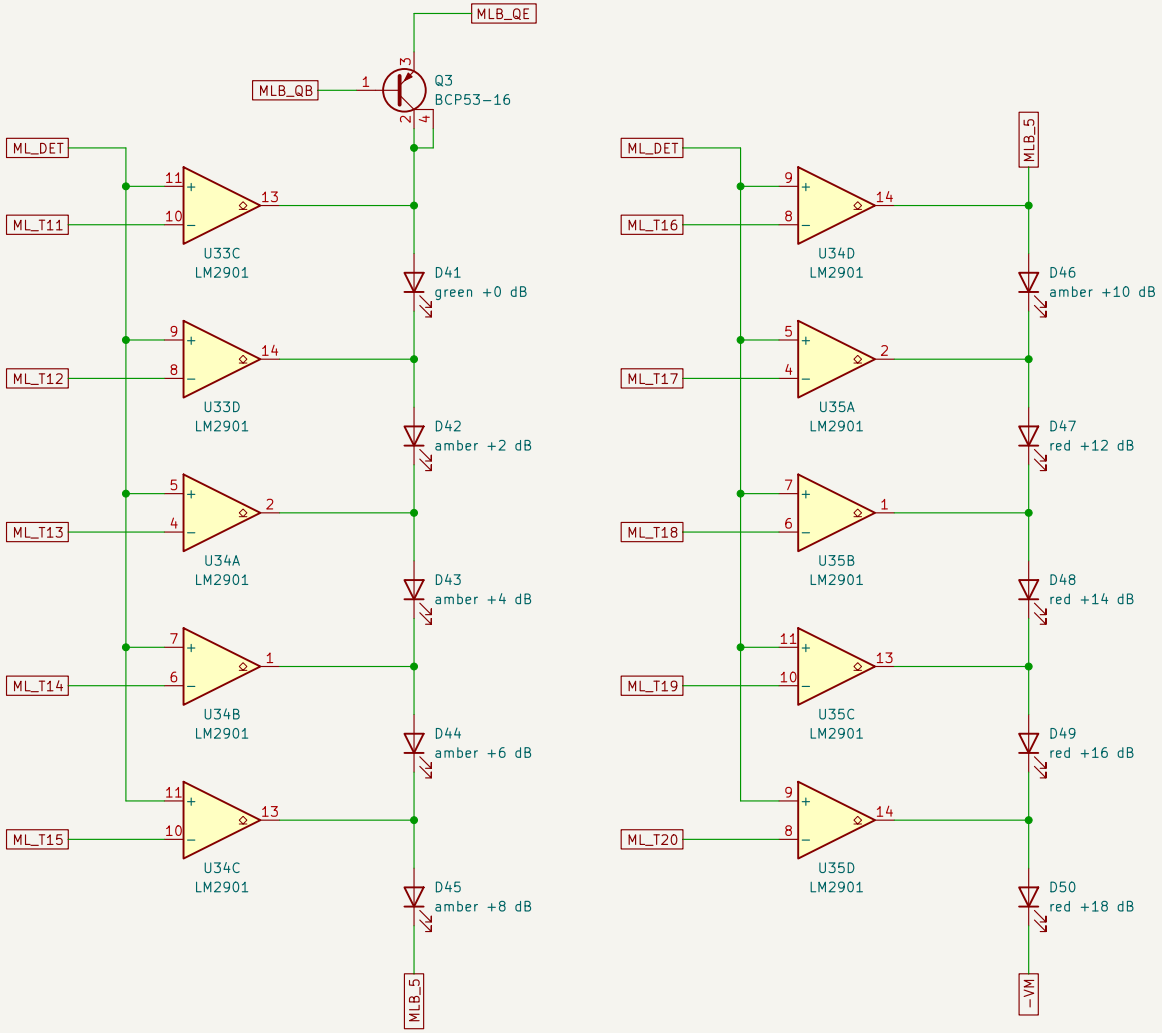
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /21 METER L/
File: 21_meter_l.kicad_sch

Title: METER L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 21/31

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



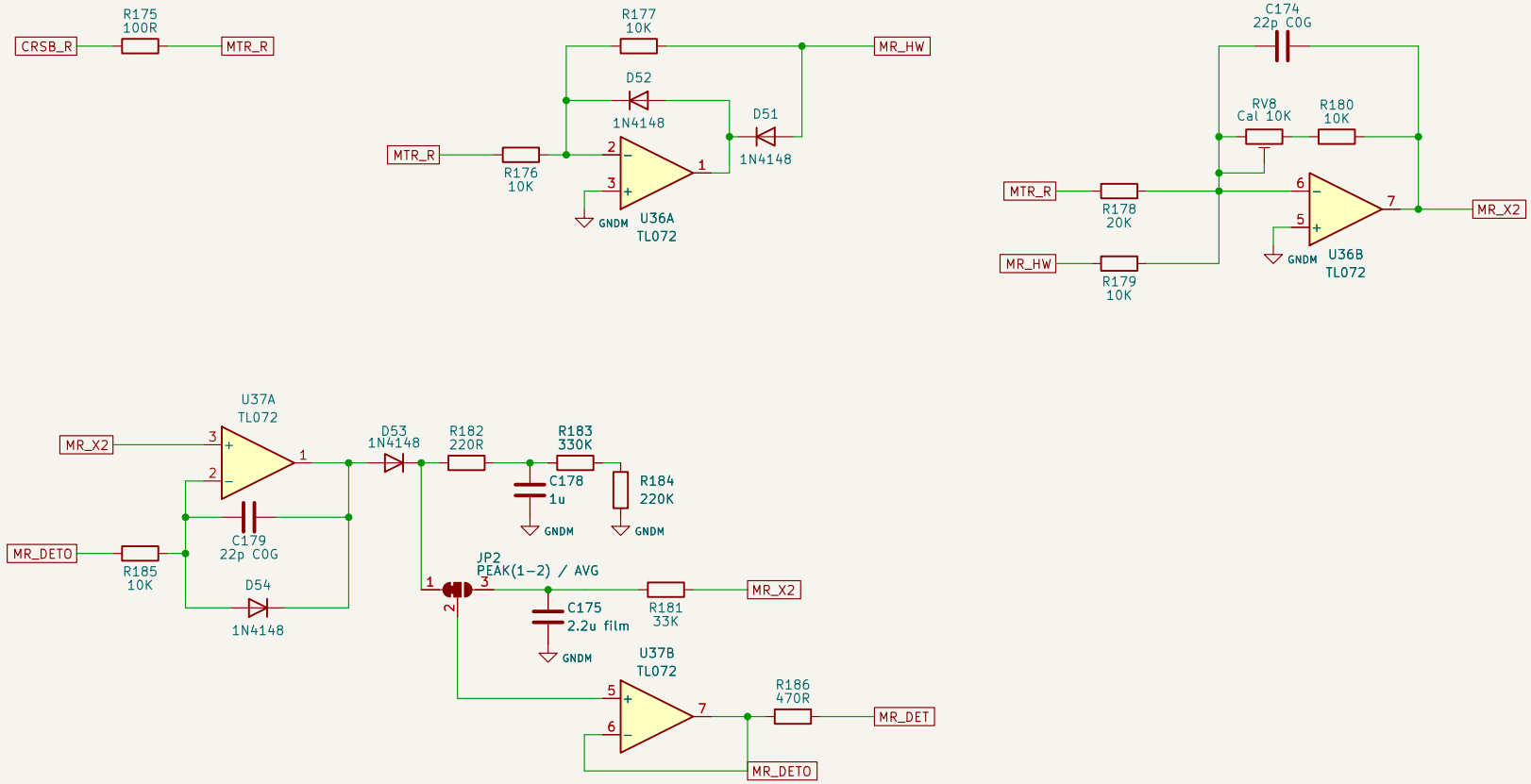
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /22 METER L/
File: 22_meter_l.kicad_sch

Title: METER L

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 22/31

METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings



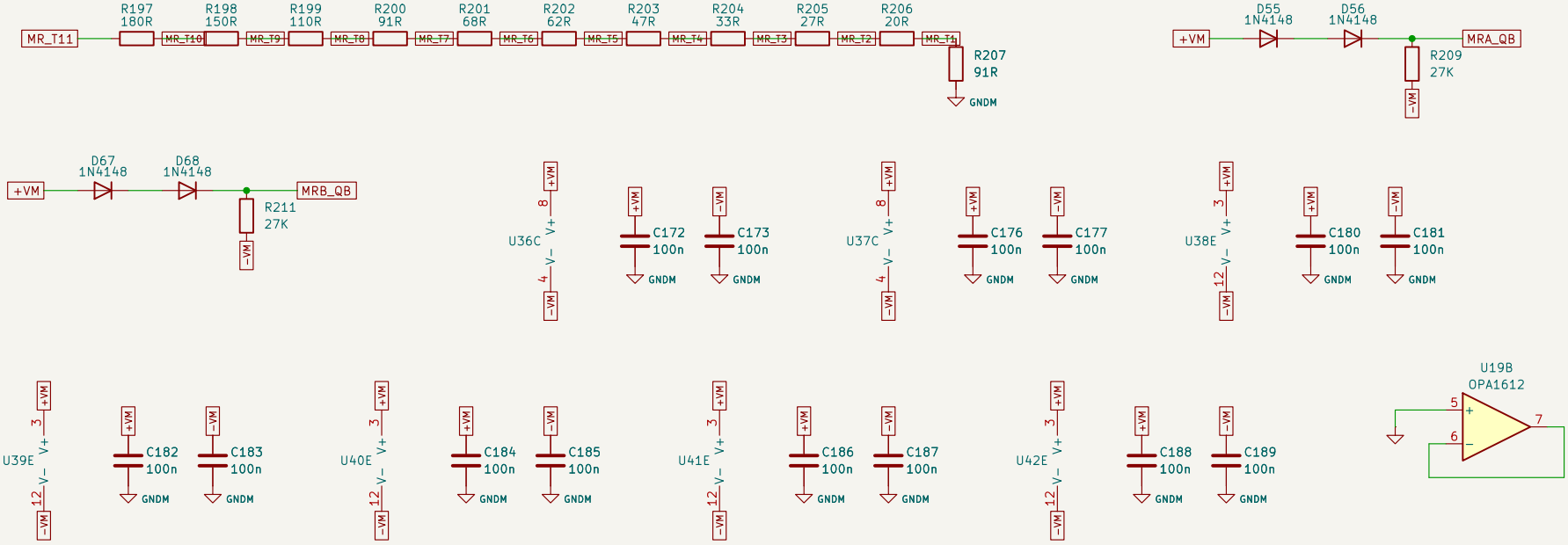
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /23 METER R/
File: 23_meter_r.kicad_sch

Title: METER R

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 23/31

METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



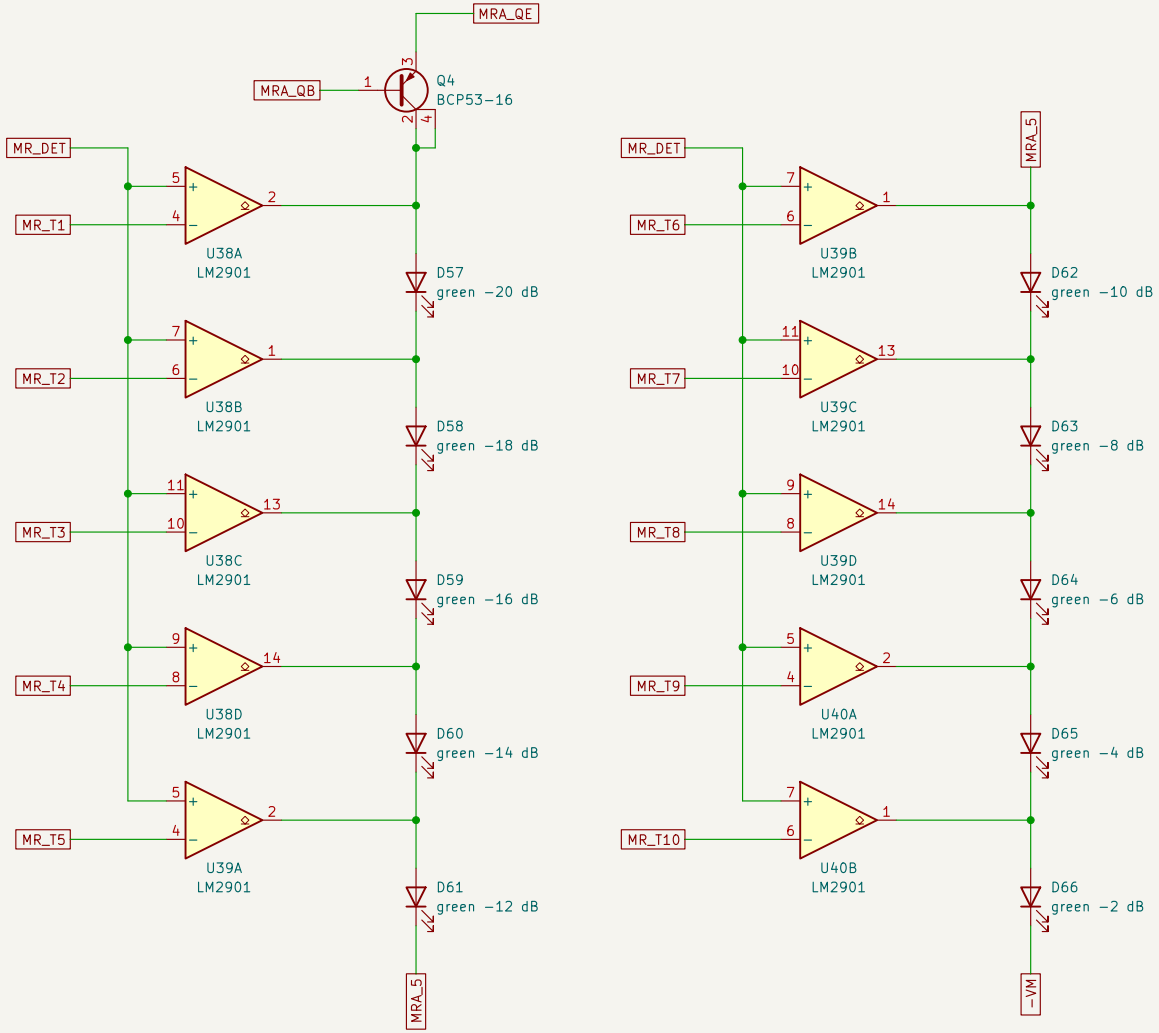
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /24 METER R/
File: 24_meter_r.kicad_sch

Title: METER R

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 24/31

METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



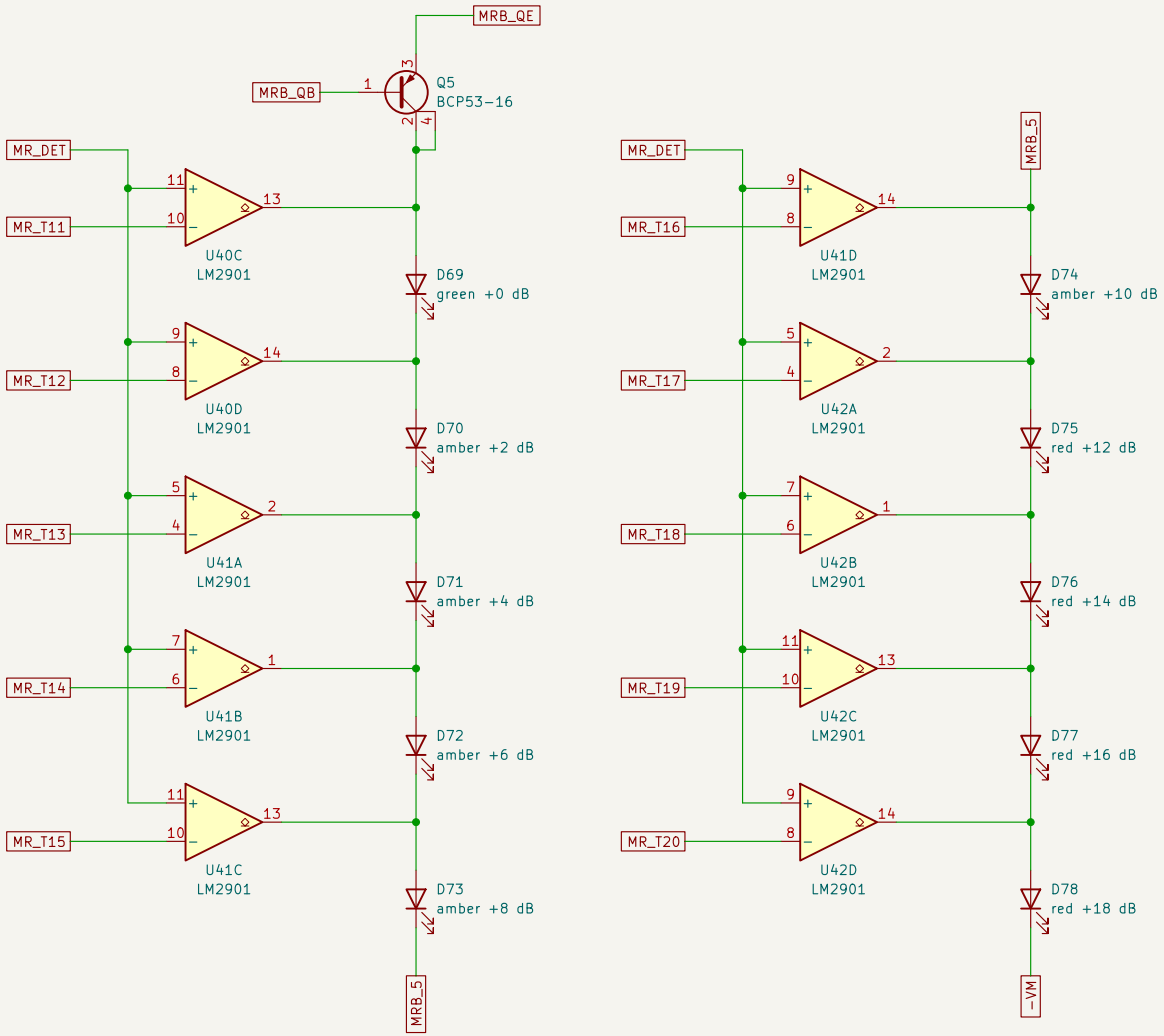
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /25 METER R/
File: 25_meter_r.kicad_sch

Title: METER R

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 25/31

METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



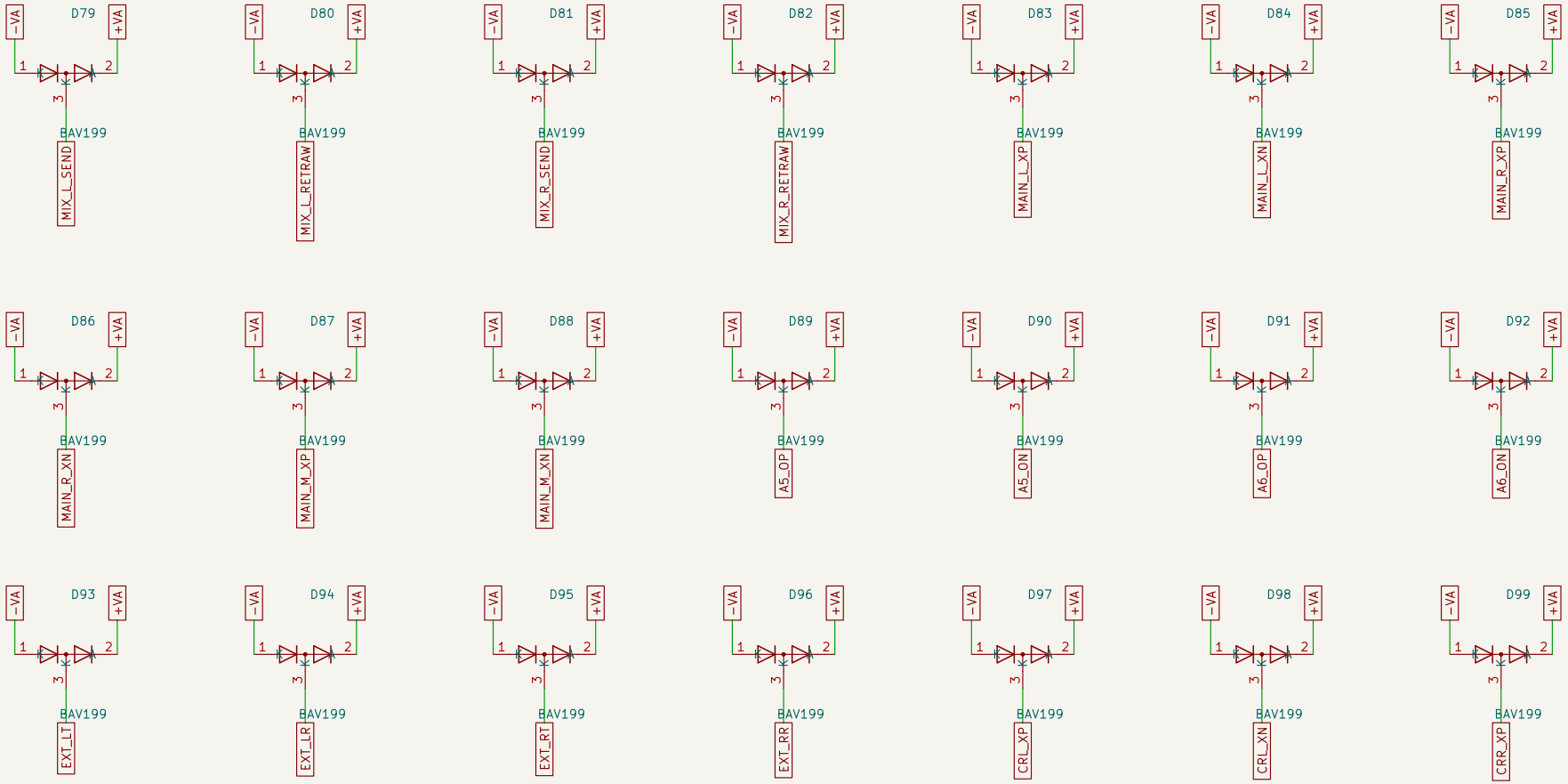
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /26 METER R/
File: 26_meter_r.kicad_sch

Title: METER R

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 26/31

JACK LINK CLAMPS: a BAV199 to the audio rails on every jack signal at the link (CS4b)



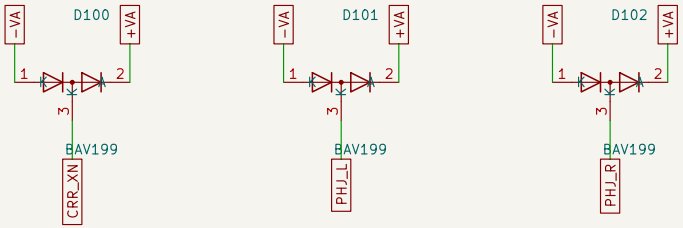
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /27 JACK LINK CLAMPS/
File: 27_jack_link_clamps.kicad_sch

Title: JACK LINK CLAMPS

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 27/31

JACK LINK CLAMPS: a BAV199 to the audio rails on every jack signal at the link (CS4b) (cont.)



SPARE BUS GROUNDING bridged solder jumpers from each spare conductor to GNDA (D13; open for SUB 5–8)



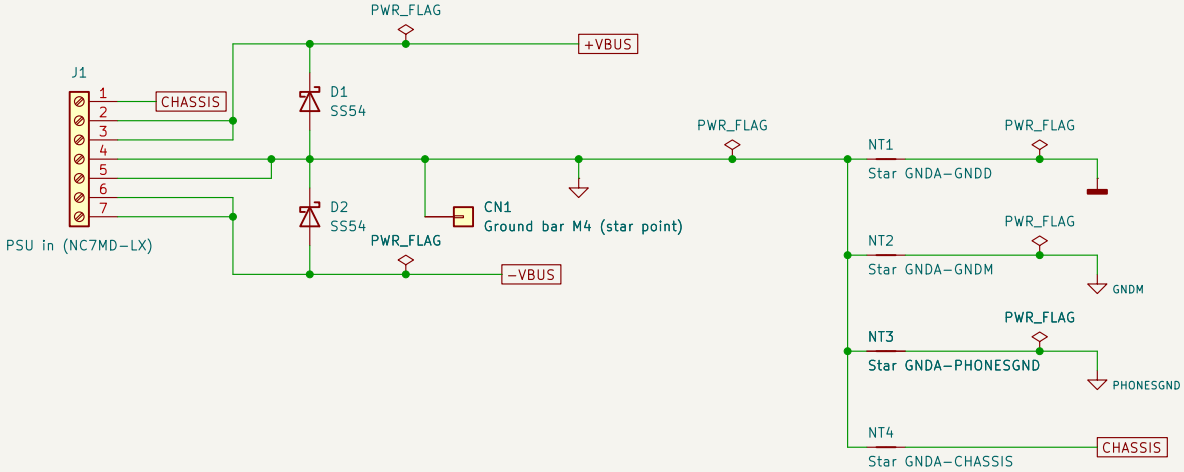
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /28 JACK LINK CLAMPS, SPARE BUS GROUNDING/
File: 28_jack_link_clamps.kicad_sch

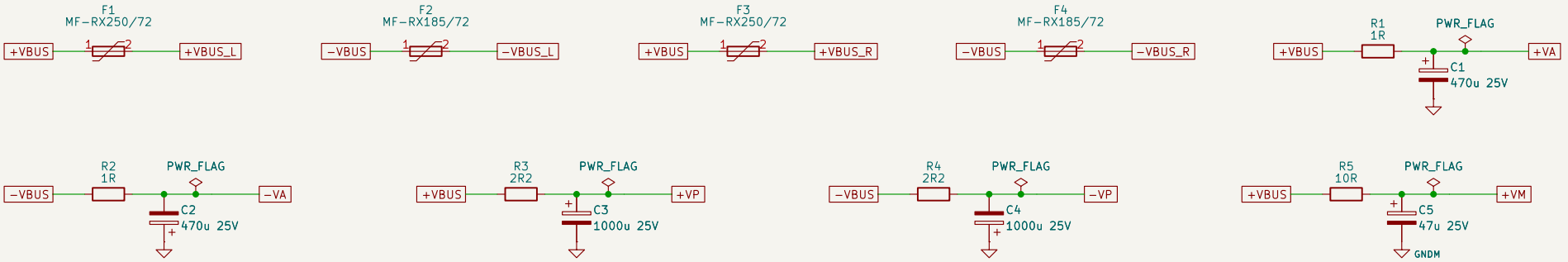
Title: JACK LINK CLAMPS, SPARE BUS GROUNDING

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 28/31

POWER ENTRY (7-pin XLR NC7MD-LX on the rear panel, two 5 A contacts per rail, 18 AWG to this terminal; +-17 V PSU)



DISTRIBUTION (PTC per wing per rail, 1.85 A hold, 72 V; local branches VA audio, VP phones, VM meters, VL logic)



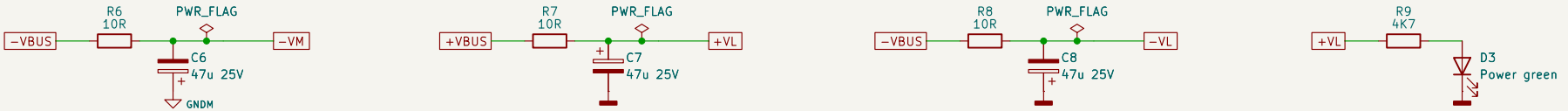
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /29 POWER ENTRY, DISTRIBUTION/
File: 29_power_entry.kicad_sch

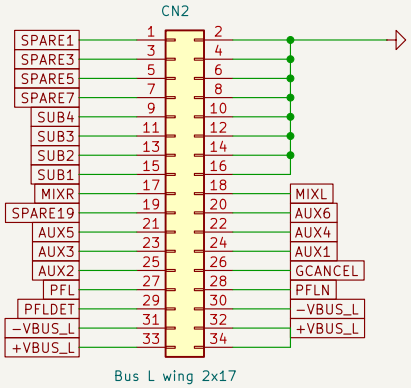
Title: POWER ENTRY, DISTRIBUTION

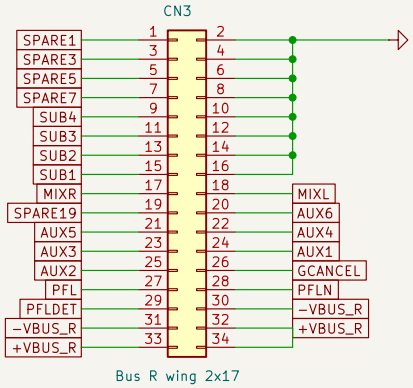
Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 29/31

DISTRIBUTION (cont.)



MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)





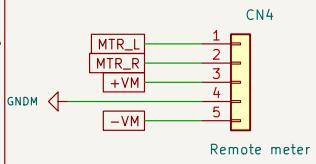
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /30 DISTRIBUTION, MOTHERLOOM/
File: 30_distribution.kicad_sch

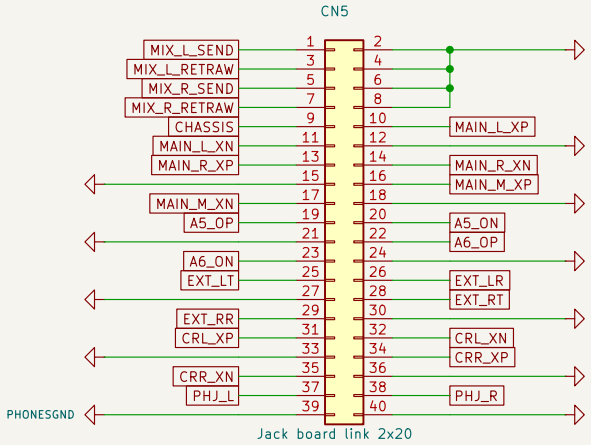
Title: DISTRIBUTION, MOTHERLOOM

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 30/31

REMOTE METER HEADER (for a meter bridge)



JACK BOARD LINK 2x20 IDC ribbon to the master jack board (9 jacks, 3 XLRs)



Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.14
Master section — goa console

Sheet: /31 REMOTE METER HEADER, JACK BOARD LINK/
File: 31_remote_meter_header.kicad_sch

Title: REMOTE METER HEADER, JACK BOARD LINK

Size: A4	Date: 2026-10-19	**Rev: 0.14**
KiCad E.D.A. 10.0.7		Id: 31/31